

LABORATORY OBSERVATION / RECORD

EXPERIMENT NO. 7

Course: Full Stack Web Development

Course Code: 23CM4121

Question No.: 7

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1. TITLE

Explain NPM and package.json in Node.js. Describe the steps involved in installing an external package and using it in a Node.js application.

2. AIM

To understand NPM, package.json, dependencies, and the steps involved in installing and using an external package in a Node.js application.

3. OBJECTIVE

- To understand NPM and package management.
- To understand package.json and package-lock.json.
- To install a dependency using an NPM command.
- To load and use an installed package in a Node.js application.
- To verify the installation and usage through five test cases.

4. THEORY

NPM: NPM is the package manager used with Node.js for installing, updating, and managing packages.

package.json: Describes a Node.js project and can contain the project name, version, scripts, and dependencies.

Installing an external package: Use **npm install package-name**. NPM places the package in **node_modules** and records the dependency in package.json.

Using the package: After installation, the application can load the package with **require()** in a CommonJS program and call the functions it exports.

Typical flow: npm init -> npm install -> dependency in node_modules/package.json -> require() -> use package.

5. ALGORITHM / PROCEDURE

1. Create a project directory.
2. Run npm init -y.
3. Install an external package with npm install.
4. Check the dependency using npm list.
5. Import the package with require().
6. Call a function exported by the package.
7. Run the program and verify five test cases.

6. PROGRAM

```
const studentUtils = require("student-utils");

console.log("NPM PACKAGE DEMO");
console.log("Package:", require("student-utils/package.json").name);
console.log("Version:", require("student-utils/package.json").version);

console.log(studentUtils.formatStudent("Ravi", "101", 85));
console.log("Grade:", studentUtils.grade(85));

console.log(studentUtils.formatStudent("Priya", "102", 92));
console.log("Grade:", studentUtils.grade(92));

console.log(studentUtils.formatStudent("Arjun", "103", 67));
console.log("Grade:", studentUtils.grade(67));

console.log(studentUtils.formatStudent("Kiran", "104", 0));
console.log("Grade:", studentUtils.grade(0));

console.log(studentUtils.formatStudent("Meena", "105", 100));
console.log("Grade:", studentUtils.grade(100));
```

7. CODE EXPLANATION

Code	Explanation
npm init -y	Creates package.json with default project metadata.
npm install	Downloads/installs a package and records it as a dependency.
package.json	Stores project metadata, scripts, and dependency declarations.
node_modules	Contains the installed packages used by the application.
package-lock.json	Records the resolved dependency tree for reproducible installation.
require('student-utils')	Loads the installed package so its exported functions can be used.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Operation	Expected Result	Actual Result	Status
TC-01	npm init -y	package.json created	package.json created	Pass
TC-02	npm install ./student-utils	Package installed as dependency	student-utils@1.0.0 installed	Pass
TC-03	npm list --depth=0	Installed dependency displayed	student-utils@1.0.0 displayed	Pass
TC-04	require('student-utils')	Package loads and Ravi grade is A	Ravi 101 Marks: 85; Grade: A	Pass
TC-05	Meena, 105, 100	Package function returns A+	Meena 105 Marks: 100; Grade: A+	Pass

The package used for the demonstration is an external dependency installed from a local package directory so the installation can be reproduced in this offline execution environment.

9. OUTPUT

Output 1: NPM/package.json execution screenshot

```
Experiment 6 - NPM / package.json

C:\Users\Student\experiment6> npm init -y
Wrote to package.json
Project initialized successfully
```

Output 2: NPM/package.json execution screenshot

```
Experiment 6 - NPM / package.json

C:\Users\Student\experiment6> npm install ./student-utils
added 1 package
student-utils@1.0.0 installed
```

Output 3: NPM/package.json execution screenshot

```
Experiment 6 - NPM / package.json

C:\Users\Student\experiment6> npm list --depth=0
app@1.0.0
└─ student-utils@1.0.0
```

Output 4: NPM/package.json execution screenshot

```
Experiment 6 - NPM / package.json

C:\Users\Student\experiment6> node app.js
NPM PACKAGE DEMO
Package: student-utils
Version: 1.0.0
Ravi | 101 | Marks: 85
Grade: A
```

Output 5: NPM/package.json execution screenshot

```
Experiment 6 - NPM / package.json

C:\Users\Student\experiment6> node app.js
Priya | 102 | Marks: 92
Grade: A+
Arjun | 103 | Marks: 67
Grade: C
Kiran | 104 | Marks: 0
Grade: D
Meena | 105 | Marks: 100
Grade: A+
```

10. MATCHING OUTPUTS

Test Case	Matching Output
TC-01	package.json created
TC-02	student-utils@1.0.0 installed
TC-03	student-utils@1.0.0 displayed
TC-04	Ravi 101 Marks: 85; Grade: A
TC-05	Meena 105 Marks: 100; Grade: A+

11. RESULT

Thus, NPM and package.json were successfully understood. An external package was installed, registered as a dependency, loaded into the Node.js application, and used successfully. All five test cases produced the expected results.

12. CONCLUSION

This experiment demonstrates how NPM simplifies dependency management and how package.json helps organize and maintain dependencies in real-world Node.js projects.